

Cache Fundamentals

Cache hit and miss

A cache hit occurs when requested data is already in the cache. A cache miss requires fetching data from the authoritative data source.

Cache key

A cache key must include the inputs that affect the result, including tenant and authorization scope. Reusing a key across tenants can expose private data.

Time to live

TTL limits how long an entry remains usable. Short TTL improves freshness but may increase database load. TTL is an expiration policy, not a guarantee of strong consistency.

Cache Write Policies

Write-through

Write-through updates the cache and authoritative storage together. It simplifies freshness but adds write latency.

Write-back

Write-back updates the cache first and writes to storage later. It improves write throughput but requires durable queues and recovery to avoid data loss.

Cache-aside

Cache-aside first reads the cache. On a miss, it fetches storage and fills the cache. A write changes storage and invalidates the cached entry.

Cache Reliability

Cache stampede

A cache stampede happens when many requests miss the same hot key simultaneously. Single-flight coalescing, expiration jitter and bounded locks reduce duplicate storage work.

Negative caching

Negative caching stores a not-found result briefly. It limits repeated requests for missing keys. Authorization failures should not be cached across users.

Measuring cache quality

Hit ratio is hits divided by total cache lookups. Measure tail latency and stale-read rate too. A high hit ratio alone does not prove correctness.